



Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

# Chapter 3.2

## Image Derivatives

*Image Processing and Computer Vision*

**LE Thanh Sach**

*Faculty of Computer Science and Engineering  
Ho Chi Minh University of Technology, VNU-HCM*

# Overview

- ➊ Image model
- ➋ First-order derivative
- ➌ Gradient
- ➍ Second-order derivative
- ➎ Laplacian
- ➏ Applications of Laplacian



Image model

First-order  
derivative

Gradient

Second-order  
derivative

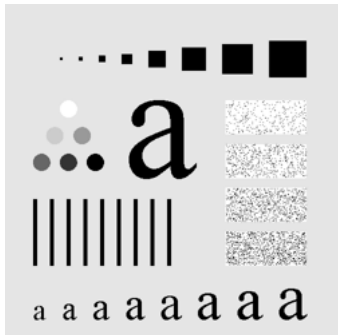
Laplacian

Applications of  
Laplacian

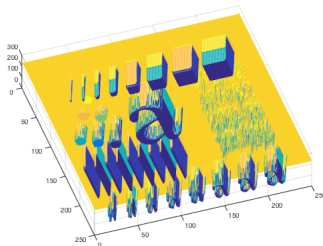
# Image Model

## Image model

- Image is a function of two variables  $x$  and  $y$ :  $f(x, y)$
- It can be seen as a surface on 2D-space.



An gray image



Mesh model of the image



## Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

# Derivative of one variable function

Taylor expansion for  $f(x + \Delta x)$ :

$$f(x + \Delta x) = f(x) + \Delta x \times f'(x) + \frac{\Delta x^2}{2!} \times f''(x) + O(\Delta x^3) \quad (1)$$

## First-order derivative from Eq. (1)

### Forward Approximation:

$$\begin{aligned} f'(x) &\cong \frac{f(x + \Delta x) - f(x)}{\Delta x} \\ &\cong f(x + 1) - f(x) \quad \text{in discrete domain} \end{aligned}$$

- This approximation has error  $O(\Delta x)$



# Derivative of one variable function

Taylor expansion for  $f(x - \Delta x)$ :

$$f(x - \Delta x) = f(x) - \Delta x \times f'(x) + \frac{\Delta x^2}{2!} \times f''(x) - O(\Delta x^3) \quad (2)$$

## First-order derivative from Eq. (2)

### Backward Approximation:

$$\begin{aligned} f'(x) &\cong \frac{f(x) - f(x - \Delta x)}{\Delta x} \\ &\cong f(x) - f(x - 1) \quad \text{in discrete domain} \end{aligned}$$

- This approximation has error  $O(\Delta x)$



## Derivative of one variable function

Eq. (1) - Eq. (2)

$$f(x + \Delta x) = f(x) + \Delta x \times f'(x) + \frac{\Delta x^2}{2!} \times f''(x) + O(\Delta x^3)$$

$$f(x - \Delta x) = f(x) - \Delta x \times f'(x) + \frac{\Delta x^2}{2!} \times f''(x) - O(\Delta x^3)$$

### First-order derivative from Eq. (1) - Eq. (2)

#### Central Approximation:

$$f'(x) \cong \frac{f(x + \Delta) - f(x - \Delta)}{2\Delta x}$$

$$\cong \frac{f(x + 1) - f(x - 1)}{2}$$

$$\cong f(x + 1) - f(x - 1)$$

**in discrete domain**

**in discrete domain**

- This approximation has error  $O(\Delta x^2)$



# Derivative of one variable function



Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

## Notation

$$f_x \equiv \frac{\partial f(x, y)}{\partial x}$$

$$f_y \equiv \frac{\partial f(x, y)}{\partial y}$$

# First-Order derivative of image



Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

| Derivatives                       | Kernel of filters                                |
|-----------------------------------|--------------------------------------------------|
| $f_x \cong f(x+1, y) - f(x, y)$   | $H_x = \begin{bmatrix} 1 & -1 \end{bmatrix}$     |
| $f_x \cong f(x, y) - f(x-1, y)$   | $H_x = \begin{bmatrix} 1 & -1 \end{bmatrix}$     |
| $f_x \cong f(x+1, y) - f(x-1, y)$ | $H_x = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}$ |



## First-Order derivative of image

In general, first-order derivatives can be computed by linear filters with the following kernels

| Derivative on x                                    | Derivative on y                                    |
|----------------------------------------------------|----------------------------------------------------|
| $H_x = \begin{bmatrix} 1 & -1 \end{bmatrix}^T$     | $H_y = \begin{bmatrix} 1 & -1 \end{bmatrix}^T$     |
| $H_x = \begin{bmatrix} -1 & 1 \end{bmatrix}^T$     | $H_y = \begin{bmatrix} -1 & 1 \end{bmatrix}^T$     |
| $H_x = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}^T$ | $H_y = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}^T$ |
| $H_x = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}^T$ | $H_y = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}^T$ |



# First-Order derivative of image

Other kernels for computing first-order derivatives

| Name    | Derivative on x                                                              | Derivative on y                                                              |
|---------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Prewitt | $H_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$ | $H_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ |
| Sobel   | $H_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$ | $H_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ |
| Robert  | $H_x = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$                        | $H_y = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$                        |



Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian



## Properties

- ① First-order derivatives are operators that can be used to obtain the variation of intensities on x and y axis.
- ② First-order derivatives can be either **positive** or **negative**.
- ③ A large variation of intensities  $\equiv$  **strong edge**  $\equiv$  large value in the absolute of derivatives.
- ④ In order to display derivatives, we need to scale derivative images.

# First-Order derivative of image

## Example

```
clear; close all; clc;
im = checkerboard(50, 5, 5);
im = uint8(255*(im > 0.5));

H_1st_x = [1 0 -1];
H_1st_y = [1 0 -1]';

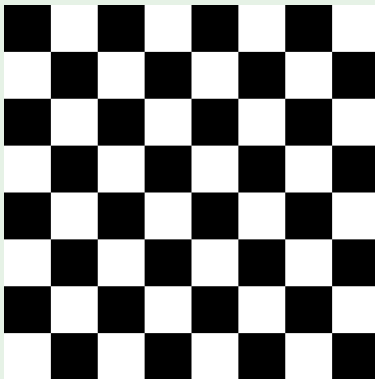
im_1st_x = abs(imfilter(double(im), H_1st_x));
im_1st_y = abs(imfilter(double(im), H_1st_y));

figure, imshow(im), title('Input_Image');
figure, imshow(gscale(im_1st_x)),
    title('1st_derivative_on_x');
figure, imshow(gscale(im_1st_y)),
    title('1st_derivative_on_y');
```



# First-Order derivative of image

## Exercise



**Figure:** a chessboard shown by `imshow(gscale(im))`

- What is the image obtained from the filtering with derivatives kernels?



# First-Order derivative of image

- Sobel and Prewitt kernel can be obtained by function **fspecial** in Matab
- In Matlab, the input image should be casted to **float** or **double** before filtering with function **imfilter**

## Example

```
clear; close all; clc;
im = imread('cameraman.tif');

H_1st_x = fspecial('Sobel');
H_1st_y = fspecial('Sobel')'; %' = transpose

im_1st_x = abs(imfilter(double(im), H_1st_x));
im_1st_y = abs(imfilter(double(im), H_1st_y));

figure, imshow(im), title('Input_Image');
figure, imshow(gscale(im_1st_x)),
title('1st_derivative_on_x');
figure, imshow(gscale(im_1st_y)),
title('1st_derivative_on_y');
```





Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

## Definition

Gradient at a pixel in a image  $f(x, y)$  is a vector  $\nabla f$ . It is defined as

$$\nabla f = \begin{bmatrix} f_x \\ f_y \end{bmatrix}$$



Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

## Magnitude and angle of gradient vectors

**Magnitude** of gradient is computed by:

$$|\nabla f| = \sqrt{f_x^2 + f_y^2}$$

or, approximated by

$$|\nabla f| = |f_x| + |f_y|$$

- Magnitude of a gradient at pixel  $(u, v)$  tells us the rate of change of intensities at  $(u, v)$
- In other words, it tells us the edge passing  $(u, v)$  is strong or not.





Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

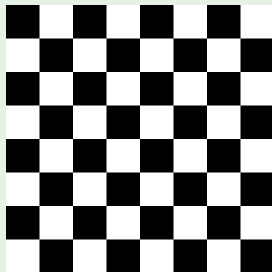
## Magnitude and angle of gradient vectors

**Angle of gradient** is computed by:

$$\theta(\nabla f) = \tan^{-1}\left(\frac{f_y}{f_x}\right)$$

- Angle of a gradient at pixel  $(u, v)$  tells us the orientation of edge passing  $(u, v)$
- Gradient vector is perpendicular to the local edge passing  $(u, v)$

## Exercise



**Figure:** a chessboard

- How do we create an image emphasizing both of horizontal and vertical edges the above chessboard?
- Which is the direction of gradient vectors at strong edge points?



## Second-order Derivative of one variable function

Second-order differential can be approximated by

$$f''(x) \cong f'(x) - f'(x+1)$$

First-order derivatives can be approximated as

$$f'(x) \cong f(x) - f(x-1)$$

$$f'(x+1) \cong f(x+1) - f(x)$$

### Second-order derivative

$$f''(x) \cong -f'(x-1) + 2f'(x) - f'(x+1)$$



# Second-Order derivative of image

## ① Second-order derivative on x

- Math:

$$\frac{\partial^2 f(x, y)}{\partial x^2} = -f(x-1, y) + 2f(x, y) - f(x+1, y)$$

- Kernel:  $H_{conv} = \begin{bmatrix} -1 & 2 & -1 \end{bmatrix}$

## ② Second-order derivative on y

- Math:

$$\frac{\partial^2 f(x, y)}{\partial y^2} = -f(x, y-1) + 2f(x, y) - f(x, y+1)$$

- Kernel:  $H_{conv} = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$



## Second-Order derivative of image

In general, second-order derivatives can be approximated by using the following kernels.

|                     | Derivative on x                                   | Derivative on y                                     |
|---------------------|---------------------------------------------------|-----------------------------------------------------|
| Positive at center: | $H_x = \begin{bmatrix} -1 & 2 & -1 \end{bmatrix}$ | $H_y = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$ |
| Negative at center: | $H_x = \begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$  | $H_y = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$  |



# First-order and Second-order derivative



Image model

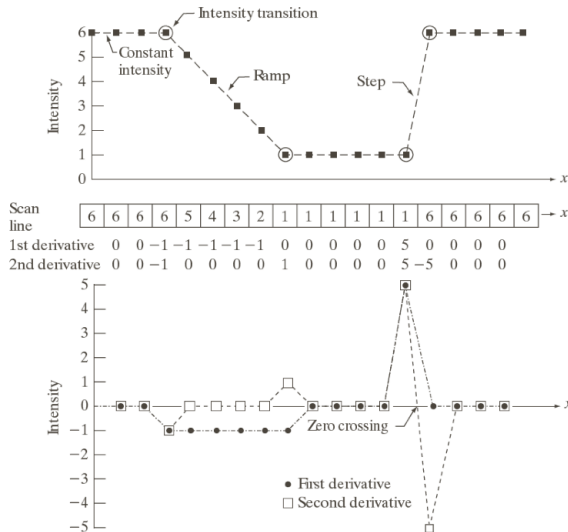
First-order derivative

Gradient

Second-order derivative

Laplacian

Applications of Laplacian



**Figure:** Example of First-order and Second-order derivatives

# First-order and Second-order derivative



## First-order derivative's properties

- ① First-order derivatives produce **thicker edge** in an image.
- ② First-order derivatives produce **strong response** to a gray-level step.

## Second-order derivative's properties

- ① Second-order derivatives produce **stronger response** to **fine detail**, such as thin lines or isolated points (noise).
- ② Second-order derivatives produce **double response** (a **positive** and a **negative**) at **step change** in in gray level.
- ③ **Zero-crossing** at a point in second-order derivatives indicates that there is an **edge passing** that point.

## Second-Order derivative of image



### Second-order derivative on x and y

Laplacian operator:

$$\nabla^2 f = \frac{\partial^2 f(x, y)}{\partial x^2} + \frac{\partial^2 f(x, y)}{\partial y^2}$$

- Math:

$$\begin{aligned}\nabla^2 f = & -f(x-1, y) + 2f(x, y) - f(x+1, y) \\ & -f(x, y-1) + 2f(x, y) - f(x, y+1)\end{aligned}$$

- Kernel:  $H_{lap} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian



## Second-Order derivative of image



### Second-order derivative on x, y, and diagonals:

Extended Laplacian operator:

- Math:

$$\begin{aligned}\nabla^2 f = & -f(x-1, y) + 2f(x, y) - f(x+1, y) \\ & -f(x, y-1) + 2f(x, y) - f(x, y+1)\end{aligned}$$

- Kernel:  $H_{lap} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$

Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

## Second-Order derivative of image

In general, Laplacian can be computed by using the following kernels. The center coefficients maybe positive or negative.

$$H_{lap} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$H_{lap} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$H_{lap} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$H_{lap} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$





Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

## Properties of Laplacian operators

- 1 **Highlight** gray-level discontinuities in images, i.e., edges on images
- 2 **De-emphasize** regions with slowly varying gray-levels

Therefore, adding (or subtracting) the original image with Laplacian image (the image obtained by filtering the original with Laplacian kernel)  $\Rightarrow$  **Sharpened images**



Method for sharpening image  $f(x, y)$ :

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) & \text{if the center coefficient of the} \\ & \text{Laplacian mask is negative} \\ f(x, y) + \nabla^2 f(x, y) & \text{if the center coefficient of the} \\ & \text{Laplacian mask is positive} \end{cases}$$

Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

## Second-Order derivative of image



Method for sharpening image  $f(x, y)$  by using linear filters.  
Kernels are as follows:

| Positive central coefficient                                                          | Positive central coefficient                                                    |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| $H_{Lap} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 9 & -1 \\ -1 & -1 & -1 \end{bmatrix}$ | $H_{Lap} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -9 & 1 \\ 1 & -1 & 1 \end{bmatrix}$ |
| $H_{Lap} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$     | $H_{Lap} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -5 & 1 \\ 0 & -1 & 0 \end{bmatrix}$ |

[Image model](#)[First-order derivative](#)[Gradient](#)[Second-order derivative](#)[Laplacian](#)[Applications of Laplacian](#)

# Applications of Laplacian operator: Examples



**Figure:** Input image: North pole of moon



Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

# Applications of Laplacian operator: Examples



Image model

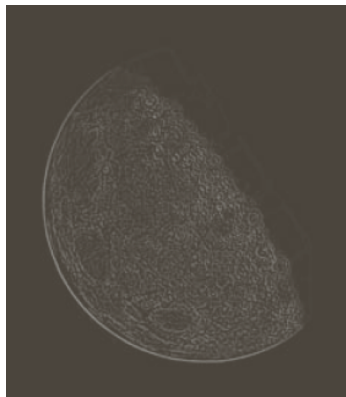
First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian



(a)



(b)

**Figure:** Laplacian images = images obtained by filtering with Laplacian operator: (a) without scaling, (b) with scaling for display purpose



## Exercise

- ① How can you sharpen images, e.g., north pole of moon, with Laplacian by using Matlab and OpenCV?
- ② Create blurred version and then sharpen the blurred version
  - Read checkboard image from file
  - Blur it with Box or Gaussian filter
  - Sharpen the resulting blurred version with Laplacian
  - Show the original, blurred version and sharpened version.

Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian



# Unsharp masking and High-boost filtering



**Unsharp masking:** image obtained from subtracting a blurred version of an image from the image itself.

$$f_s(x, y) = f(x, y) - \bar{f}(x, y)$$

- $\bar{f}(x, y)$  : blurred version of  $f(x, y)$ , e.g., using box filter
- $f_s(x, y)$ : sharpened image



## Exercise

- ① How can you implement unsharp masking with Matlab and OpenCV?
- ② Create blurred version and then sharpen the blurred version
  - Read checkboard image from file
  - Blur it with Box or Gaussian filter
  - Sharpen the resulting blurred version with Unsharp masking
  - Show the original, blurred version and sharpened version.

Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian



**High-boost filtering:** a generalized unsharpened image.

$$f_{hb}(x, y) = Af(x, y) - \bar{f}(x, y)$$

- $A$  : a parameter,  $A \geq 1$ .  $A = 1$ , high-boost filtering  $\rightarrow$  unsharp masking
- $f_s(x, y)$ : sharpened image

Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian



**High-boost filtering:** Other representations.

$$\begin{aligned}f_{hb}(x, y) &= Af(x, y) - \bar{f}(x, y) \\&= (A - 1)f(x, y) + f(x, y) - \bar{f}(x, y) \\&= (A - 1)f(x, y) + f_s(x, y)\end{aligned}\quad (3)$$

Replace Eq. (3) with Laplacian sharpening filter:

$$f_{hb}(x, y) = \begin{cases} Af(x, y) - \nabla^2 f(x, y) & \text{if the center coefficient of the} \\ & \text{Laplacian mask is negative} \\ Af(x, y) + \nabla^2 f(x, y) & \text{if the center coefficient of the} \\ & \text{Laplacian mask is positive} \end{cases}$$

Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

## Unsharp masking and High-boost filtering



**High-boost filtering:** can be implemented as a linear filter, with one of the following kernels.

$$H_{hb8} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & A+8 & -1 \\ -1 & -1 & -1 \end{bmatrix} \quad \parallel \quad H_{hb4} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & A+4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

**A is a parameter:**

- 1  $A = 0$ , high-boost filter becomes standard **Laplacian filter**
- 2  $A = 1$ , high-boost filter becomes standard **Laplacian sharpening filter**
- 3  $A$  increases pass 1 the contribution of sharpening process becomes less and less important

# Unsharp masking and High-boost filtering



Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian

## Applications of High-boost filtering:

- High-boost filtering can sharpen images
- High-boost filtering can brighten dark images

# Unsharp masking and High-boost filtering: Examples



a  
b  
c  
d  
e

**FIGURE 3.40**

- (a) Original image.  
(b) Result of blurring with a Gaussian filter.  
(c) Unsharp mask. (d) Result of using unsharp masking.  
(e) Result of using highboost filtering.



Image model

First-order  
derivative

Gradient

Second-order  
derivative

Laplacian

Applications of  
Laplacian